Tanmay Bhuskute

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SUMMARY

M.S. Computer Science candidate (Dec 2026) and Software Engineer with hands-on PyTorch, TensorFlow, and OpenCV experience building and evaluating vision/ML pipelines, plus production engineering background in Python, AWS, and Kubernetes. Current research evaluates vision-transformer and object-detection models (CLIP, GroundingDINO) for real-time automation.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Deep Learning, Advanced Algorithms, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Machine Learning, Data Mining, Data Analytics, Object-Oriented Programming.

TECHNICAL SKILLS

Vision & ML: PyTorch, TensorFlow, OpenCV, scikit-learn, Vision Transformers (CLIP), convolutional/object-detection models (GroundingDINO), model quantization, NumPy, pandas.

Programming: Python (OOP, decorators, context managers), Java, JavaScript, TypeScript, C/C++, SQL, Go.

Web & Backend: React, Node.js, Django, Flask, FastAPI, REST APIs.

Databases: PostgreSQL, MySQL.

Developer Tools: Git, GitHub, GitLab CI/CD, Jenkins, AWS, Docker, Kubernetes, Terraform, Linux, Bash.

WORK EXPERIENCE

Graduate Researcher —- University of Cincinnati Feb 2026 – Present

- Evaluated vision-model components for real-time automation, benchmarking a **GPT-4o + GroundingDINO (object detection) + CLIP (vision transformer)** pipeline against a proposed memory-augmented LLM method across 53 runs on 20+ Android apps.
- Built PyTorch/Python evaluation harness measuring inference time and task-completion fidelity, demonstrating **25.7x latency reduction** (39.4s vs 1012.1s avg per run) relevant to real-time edge deployment constraints.

Software Development Engineer —- Siemens Industry Software Jul 2022 – Jun 2024

- Automated **Teamcenter X** infrastructure workflows with Python (Boto3, OOP-driven modular design), replacing legacy Ruby scripts and reducing deployment time by **40%** across AWS environments.
- Executed **44 production environment upgrades** in 4 months, diagnosing live infrastructure issues across Linux, EC2, databases, and Teamcenter X services.
- Designed **Ansible-based Blue-Green deployment** framework, reducing production upgrade downtime by **50%**.
- Migrated **40+ production environments** to Kubernetes using ArgoCD, Helm, and GitLab CI/CD, collaborating via GitHub with clear, well-structured documentation.
- Served as Quality Coach and SonarQube Specialist, increasing automated code coverage from **35% to 78%**; mentored **5 software engineers** through structured onboarding.

Software Developer Intern —- Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** framework, catching **95%** of misconfigurations pre-production.

PROJECTS AND RESEARCH

NYC Mobility Forecasting [Python, PyTorch, scikit-learn, PySpark, FastAPI] Jul 2026

- Built ML system forecasting NYC taxi/rideshare demand across **263 zones** (**2.3M-row** hourly dataset); designed two-stage Poisson hurdle model improving MAPE **18.77% to 16.52%** and cutting RMSE **43%** over baseline Random Forest.

News Headline Classification – Deep Learning NLP Model [Python, PyTorch, spaCy] Mar 2025

- Built LSTM-based PyTorch classifier on **400K+** headlines, achieving **93.28% accuracy**; optimized preprocessing/training pipeline (NumPy, pandas) improving efficiency by **25%**.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).